



Maintenance Schedule

Diesel engine

12V4000G73 / T1238A34-2150

16V4000G73 / T1638A34-2152

Application group 3A

MS50009/04E

12V4000G73
16V4000G73

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL 1 to QL4.

QL 1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Load profile

Load profile

Power	110%	100%
Corresponding operating time	1%	99%

1.4 Maintenance schedule matrix

0 to 14,500 operating hours

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
Engine																															
W1024	1 w																														
W0500	-	X																													
W0501	-	X																													
W0503	-	X																													
W0505	-	X																													
W0506	-	X																													
W0525	-	X																													
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1864	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1008	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1247	3 m			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1463	1 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a							X						X						X						X					
W1005	3 a							X						X						X						X					
W1207	18 a			X				X						X						X						X					
W1689	18 a							X						X						X						X					
W1006	18 a													X												X					
W1713	18 a													X												X					
W1011	18 a													X												X					
W1046	18 a																														
W1026	5 a																														
W1812	5 a																														
W1250	6 a																														
W2000	18 a																														
W2001	18 a																														
W2004	18 a																														
W2006	18 a																														
W2007	18 a																														
W2008	18 a																														
W2009	18 a																														
W2010	18 a																														
W2062	18 a																														
W2070	18 a																														
W2110	18 a																														
W1041	18 a																														
W1058	18 a																														
W1251	6 a																														
W1134	18 a																														
W3000	18 a																														
W3001	18 a																														
w = weeks m = months a = years																															

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
W3002	18 a																														
W3003	18 a																														
W3004	18 a																														
W3005	18 a																														
W3010	18 a																														
W3018	18 a																														
W3021	18 a																														
W3042	18 a																														
W3083	18 a																														
W3100	18 a																														
W3103	18 a																														
W3108	18 a																														
W3177	18 a																														
W3182	18 a																														
W1124	18 a																														
W1878	18 a																														
W1478	18 a																														
w = weeks m = months a = years																															

15,000 to 29,500 operating hours

Task	Limit	Operating hours [h]																													
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000	24,500	25,000	25,500	26,000	26,500	27,000	27,500	28,000	28,500	29,000	29,500
Engine																															
W1024	1 w																														
W0500	-																														
W0501	-																														
W0503	-																														
W0505	-																														
W0506	-																														
W0525	-																														
W1009	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1864	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1008	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1247	3 m	X		X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1463	1 a	X		X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a	X		X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1001	2 a	X		X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a	X						X						X						X						X					
W1005	3 a	X						X						X						X						X					
W1207	18 a	X						X						X						X						X					
W1689	18 a	X						X						X						X						X					
W1006	18 a							X												X											
W1713	18 a							X												X											
W1011	18 a							X												X											
w = weeks m = months a = years																															

TIM-ID: 000.0082961 - 001

Task	Limit	Operating hours [h]																			
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000	24,500
W1046	18 a	X																			
W1026	5 a	X																			
W1812	5 a	X																			
W1250	6 a	X																			
W2000	18 a	X																			
W2001	18 a	X																			
W2004	18 a	X																			
W2006	18 a	X																			
W2007	18 a	X																			
W2008	18 a	X																			
W2009	18 a	X																			
W2010	18 a	X																			
W2062	18 a	X																			
W2070	18 a	X																			
W2110	18 a	X																			
W1041	18 a	X																			
W1058	18 a	X																			
W1251	6 a																				
W1134	18 a																				
W3000	18 a																				
W3001	18 a																				
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W3003	18 a																				
W3004	18 a																				
W3005	18 a																				
W3010	18 a																				
W3018	18 a																				
W3021	18 a																				
W3042	18 a																				
W3083	18 a																				
W3100	18 a																				
W3103	18 a																				
W3108	18 a																				
W3177	18 a																				
W3182	18 a																				
W1124	18 a																				
W1878	18 a																				
W1478	18 a																				
w = weeks m = months a = years																					

30,000 to 30,000 operating hours

Task	Limit	Operating hours [h]																											
		30,000																											
Engine																													
W1024	1 w																												
W0500	-																												
W0501	-																												
W0503	-																												
W0505	-																												
W0506	-																												
W0525	-																												
W1009	2 a	X																											
W1864	2 a	X																											
W1008	2 a	X																											
W1247	3 m	X																											
W1463	1 a	X																											
W1575	1 a	X																											
W1001	2 a	X																											
W1241	2 a	X																											
W1005	3 a	X																											
W1207	18 a	X																											
W1689	18 a	X																											
W1006	18 a	X																											
W1713	18 a	X																											
W1011	18 a	X																											
W1046	18 a	X																											
W1026	5 a	X																											
W1812	5 a	X																											
W1250	6 a	X																											
W2000	18 a	X																											
W2001	18 a	X																											
W2004	18 a	X																											
W2006	18 a	X																											
W2007	18 a	X																											
W2008	18 a	X																											
W2009	18 a	X																											
W2010	18 a	X																											
W2062	18 a	X																											
W2070	18 a	X																											
W2110	18 a	X																											
W1041	18 a	X																											
W1058	18 a	X																											
W1251	6 a	X																											
W1134	18 a	X																											
W3000	18 a	X																											
W3001	18 a	X																											
W3002	18 a	X																											
W3003	18 a	X																											
w = weeks m = months a = years																													

Task	Limit	Operating hours [h]																													
		30,000																													
W3004	18 a	X																													
W3005	18 a	X																													
W3010	18 a	X																													
W3018	18 a	X																													
W3021	18 a	X																													
W3042	18 a	X																													
W3083	18 a	X																													
W3100	18 a	X																													
W3103	18 a	X																													
W3108	18 a	X																													
W3177	18 a	X																													
W3182	18 a	X																													
W1124	18 a	X																													
W1878	18 a	X																													
W1478	18 a	X																													
w = weeks m = months a = years																															